

The Jet Properties and Accretion Regime for Blazar Sequence

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ABSTRACT

The puzzling bimodality displayed by blazars on the broad-band spectral indices ($\alpha_{\text{ro}} - \alpha_{\text{ox}}$) plane is still an open question. To investigate its physical origin, we compiled a sample comprising 136 flat-spectrum radio quasars (FSRQs), 64 low-synchrotron-peaked BL Lac objects (LBLs), 39 intermediate-synchrotron-peaked BL Lac objects (IBLs), and 105 high-synchrotron-peaked BL Lac objects (HBLs). Our results show that (FSRQs+LBLs)→IBLs→HBLs follow a $\searrow$ -shaped evolutionary sequence on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane, with the average 5 GHz radio Doppler factor, jet power, and accretion Eddington-ratio decreasing along this $\searrow$ -shaped track. Additionally, the distributions of α_{ox} exhibit only slight differences among blazar subclasses, whereas those of α_{ro} and α_{rx} differ significantly. Our results can be naturally explained if the distribution of blazars on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane is regulated by the accretion rates and the Doppler beaming effect across different bands ($\delta_i^{q_i}/\delta_j^{q_j}$). However, δ_i/δ_j plays a pivotal role in the distribution, while jet shape ($q = 2(3) + \alpha_\nu$) and spectral indices (α_ν) play a minor role. Our results suggest that FSRQs+LBLs are strong-jet sources coupled with radiatively efficient accretion, and their jets are accelerated to a larger distance from the core and still maintain relativistic speeds down to the radio region; HBLs are weak-jet sources coupled with radiatively inefficient accretion, and their jets may begin to decelerate in the optical region and exhibit sub-relativistic speeds as they propagate to the radio region. IBLs are regarded as transitional objects between FSRQs+LBLs and HBLs.

Keywords: Black Hole Physics – Galaxy Nuclei — Accretion disk — Relativistic jets — Active galactic nuclei — Blazars — Flat-spectrum radio quasars — BL Lacertae objects

1. INTRODUCTION

Blazars are a subclass of active galactic nuclei (AGNs) with a relativistic jet aligned with our line of sight (C. M. Urry & P. Padovani 1995), which causes the broad-band emissions of blazars to be boosted and confers them with many extreme properties (e.g., super-luminal motions, rapid variability, high and variable polarization, high luminosity, and high energy γ -ray emissions, see B. J. Wills et al. 1992; J. H. Fan & G. Z. Xie 1996; J. H. Fan & R. G. Lin 2000; J.-H. Fan 2005; J.-H. Fan et al. 2013; C. M. Raiteri et al. 2017; S. Abdollahi et al. 2022). Blazars consist of two subclasses, namely flat-spectrum radio quasars (FSRQs) and BL Lac objects (BL Lacs), distinguished by the equivalent width (EW) of their optical emission lines. FSRQs exhibit strong broad emission lines with $\text{EW} > 5\text{\AA}$, while BL Lacs have no or weak emission lines characterized by $\text{EW} < 5\text{\AA}$ (R. Scarpa & R. Falomo 1997). The discrepancy of emission lines between FSRQs and BL Lacs was explained by the different accretion regimes or the Doppler beaming effect (e.g., E. A. Corbett et al. 2000; J. H. Fan 2003; G. Ghisellini & F. Tavecchio 2008; G. Ghisellini et al. 2011; P. Padovani et al. 2012; L. Foschini 2012; T. Sbarrato et al. 2014; G. Chen et al. 2025).

The typical feature of blazars is a double-hump structure in the spectral energy distribution (SED, see G. Fossati et al. 1998; D. Donato et al. 2001; J. H. Fan et al. 2016; R. Blandford et al. 2019; J. H. Yang et al. 2022). The lower-energy hump, which spans from radio to X-ray band, is characterized by synchrotron radiation; the higher-energy

hump is located at MeV to TeV band (G. Fossati et al. 1998; E. Nieppola et al. 2006; J. H. Fan et al. 2016). However, the origin of the high-energy bump remains controversial. It may arise either from inverse Compton scattering of soft photons by relativistic electrons (the leptonic model) or from synchrotron emission by ultrarelativistic protons and hadronic cascades (the hadronic model; see M. Böttcher et al. 2013; M. Cerruti et al. 2015). Based on the location of peak frequency (ν_p^{syn}) of synchrotron hump in SED, BL Lacs are divided into low-synchrotron-peaked BL Lacs (LBLs: $\nu_p^{\text{syn}} < 10^{14}$ Hz), intermediate-synchrotron-peaked BL Lacs (IBLs: 10^{14} Hz $< \nu_p^{\text{syn}} < 10^{15}$ Hz), and high-synchrotron-peaked BL Lacs (HBLs: $\nu_p^{\text{syn}} > 10^{15}$ Hz, see A. A. Abdo et al. 2010). FSRQs are usually powerful sources with $\nu_p^{\text{syn}} \lesssim 10^{14}$ Hz (A. A. Abdo et al. 2010; P. Giommi et al. 2012; J. H. Fan et al. 2016; J. H. Yang et al. 2022). BL Lacs are classified as radio-selected BL Lacs (RBLs) and X-ray-selected BL Lacs (XBLs) based on the surveys that discovered them; RBLs tend to be LBLs, and XBLs tend to be HBLs (P. Giommi et al. 1995; Z. Wu et al. 2007). Additionally, BL Lacs are also divided into LBLs and HBLs according to the broad-band spectral index (also known as the effective spectral index) from 5 GHz radio band to 1 keV X-ray band (α_{rx} , the broad-band spectral indices α_{ij} are defined in the following as Eq (1), see J. E. Ledden & S. L. O’Dell 1985). LBLs have $\alpha_{\text{rx}} > 0.75$, while HBLs have $\alpha_{\text{rx}} < 0.75$ (P. Padovani & P. Giommi 1995; J. M. Bai & M. G. Lee 2001).

The above-mentioned differences in observational properties among blazar subclasses suggest that the underlying physics differs among them. The differences in SED likely suggest different accretion regimes (M. Böttcher & C. D. Dermer 2002; J.-M. Wang et al. 2002; G. Ghisellini & F. Tavecchio 2008; V. S. Paliya et al. 2021). M. Keenan et al. (2021) suggested that FSRQ, LBLs, and IBLs are strong-jet sources associated with the radiatively efficient accretion, while HBLs are weak-jet sources that exhibit the radiatively inefficient advection-dominated accretion flows (ADAF, see also E. T. Meyer et al. 2011; X.-H. Ye et al. 2025). Furthermore, the differences in relativistic jet effect among blazars have also been found, namely, an anti-correlation between the radio Doppler factor (δ_r) and ν_p^{syn} (e.g., Z. Wu et al. 2007; E. Nieppola et al. 2008; W. X. Yang et al. 2022; Q.-C. Long et al. 2025), which implies the different jet properties among the blazar subclasses. Interestingly, Very Long Baseline Array (VLBA) observations on parsec scales show that the radio knots in FSRQs exhibit super-luminal motions, LBLs exhibit sub- to super-luminal motions, whereas the radio components in HBLs generally move slowly and, in some cases, remain stationary (e.g., S. G. Jorstad et al. 2001, 2017; B. G. Piner et al. 2008, 2010; M. L. Lister et al. 2009, 2013; P. Thevenet et al. 2026). Obviously, VLBA observations are consistent with the anti-correlation of $\delta_r - \nu_p^{\text{syn}}$. Observational studies across different blazar subclasses seem to support an emerging consensus that strong-jet sources are associated with accelerating jets, whereas weak-jet sources are associated with decelerating jets (e.g., J. H. Fan et al. 1994; M. Georganopoulos & D. Kazanas 2003; G. Ghisellini et al. 2005; B. G. Piner et al. 2008, 2010; V. Karamanavis et al. 2016; S. G. Jorstad et al. 2017; I. Liodakis et al. 2018; Q.-C. Long et al. 2025, 2026).

A crucial way of understanding the blazar physics is to study their SED behaviour, which can be parameterized using the broad-band spectral indices (B. Balmaverde et al. 2006). P. Padovani & P. Giommi (1995) found that LBLs and HBLs dwell in two completely different regions of the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane (see also J. E. Ledden & S. L. O’Dell 1985; D. Donato et al. 2001; B. Balmaverde et al. 2006; E. Nieppola et al. 2006; A. A. Abdo et al. 2010; J. H. Fan et al. 2016; J. H. Yang et al. 2022, where the broad-band frequencies are usually defined at the 5 GHz radio band, the optical V-band, and the 1 keV X-ray band). Most LBLs occupy the upper right region of the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane with $\alpha_{\text{rx}} > 0.75$, most HBLs occupy the lower left corner of the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane with $\alpha_{\text{rx}} < 0.75$, IBLs appear to serve as a bridge between LBLs and HBLs (E. Nieppola et al. 2006). Interestingly, FSRQs occupy the region of LBLs on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane (D. Donato et al. 2001; A. A. Abdo et al. 2010; J. H. Fan et al. 2016; J. H. Yang et al. 2022).

The puzzling bimodality displayed by blazars on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane indicates their different energy distributions in radio, optical, and X-ray bands, which was explained as the result of migration of ν_p^{syn} (see Figure 12 of P. Padovani & P. Giommi 1995). In the unified model of AGNs, the discrepancy of observational properties is regulated by accretion and orientation (Y. Shen & L. C. Ho 2014). However, the jet orientation makes it difficult for any observational properties of blazars to avoid the impact of Doppler beaming effect. Therefore, the observational properties of blazars should be related to the accretion and the beaming effect. Several early studies have suggested that the beaming effect could affect the distribution of sources on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane (see J. H. Fan & G. Z. Xie 1996; M. Chiaberge et al. 2000; A. Capetti et al. 2000; E. Trussoni et al. 2003; B. Balmaverde et al. 2006). However, this proposal remains a theoretical prediction, and direct observational evidence has yet to be obtained. In this paper, we aim to (1) verify whether the beaming effect impacts the distribution of blazars on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane and investigate how the beaming effect influences the distribution of blazars, (2) explore accretions and jet physics behind this phenomenological bimodality.

The structure of the paper is as follows. In Section 2, we describe our blazars sample in detail, calculate the broad-band spectral indices and some parameters. In Section 3, we present our results. The discussions and summaries of our results are presented in Sections 4 and 5, respectively. Throughout this paper, we adopt a flat- Λ CDM cosmology with $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_\Lambda = 0.73$, $\Omega_M = 0.27$, and the web-based cosmology calculator (E. L. Wright 2006).

2. BLAZAR SAMPLE

To achieve the objectives of this work, our sample includes all subclasses of blazars. In addition, the selected blazars were required to satisfy the following criteria: (1) Available measurements of ν_p^{syn} , allowing us to classify the sources. (2) Available radio, optical, and X-ray fluxes, allowing us to calculate the broad-band spectral indices. (3) Available Doppler factors (δ_ν), allowing us to investigate whether Doppler beaming affects the distribution of blazars on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane. (4) Available low-frequency radio fluxes, allowing us to estimate the intrinsic jet power ($P_{\text{jet}}^{\text{in}}$). (5) Available accretion luminosity (L_{disk}) and dynamical black hole (BH) mass (M_{BH}), allowing us to estimate the accretion rates (it is quantified by the ratio of L_{disk} to the Eddington luminosity (L_{Edd}): $\lambda_{\text{Edd}} = \log(L_{\text{disk}}/L_{\text{Edd}})$). However, most BL Lacs have no available L_{disk} or M_{BH} . To obtain as large as possible a blazar sample, blazars that meet the first four criteria are included.

The third data release of the Fourth Catalog of AGNs Detected by *Fermi*-LAT (4LAC-DR3, M. Ajello et al. 2022) provides ν_p^{syn} measurements for sources in 4FGL catalog (Although a few blazars do not have available ν_p^{syn} in 4LAC-DR3, we find that their ν_p^{syn} are available in E. Nieppola et al. 2006; D. Xiong et al. 2015a; Y.-L. Chang et al. 2019). Several studies have estimated the δ_r for blazars (δ_r -catalogs, see T. Hovatta et al. 2009; Z. Wu et al. 2014; I. Liodakis et al. 2017, 2018; X.-H. Ye & J.-H. Fan 2021). We therefore cross-matched 4LAC-DR3 with these δ_r -catalogs and further examine whether the matched sources have available broad-band data. In total, we constructed a sample containing 136 FSRQs, 64 LBLs, 39 IBLs, and 105 HBLs that have available ν_p^{syn} , 5 GHz core radio flux, optical flux, X-ray flux, δ_r , and low frequency radio flux. Most of our blazars (281 blazars, see Table 3) have available M_{BH} .

2.1. The Collection and Analysis of Data

Most of the ν_p^{syn} of our blazars are taken from M. Ajello et al. (2022), only a few from E. Nieppola et al. (2006), D. Xiong et al. (2015b), and Y.-L. Chang et al. (2019).

To mitigate the possible contamination from the extended emissions (F. Panessa et al. 2019), the 5 GHz core radio flux (f_r) is adopted in our analysis. The available f_r are extracted from NASA/IPAC Extragalactic Database (NED⁴) and the literatures (see table 1). In this work, the optical emissions at the $5.4143 \times 10^{14} \text{ Hz}$ V-band are adopted, the observational optical fluxes of our blazars are taken from the SIMBAD database, operated at CDS Strasbourg, France (CDS⁵). In line with previous works (J. E. Ledden & S. L. O'Dell 1985; J. H. Fan et al. 2016), we use the observational 1 keV X-ray flux ($f_{1\text{keV}}$) for our analysis. However, only a few of blazars have available $f_{1\text{keV}}$. Therefore, for most of our blazars, $f_{1\text{keV}}$ are extrapolated from other approximate bands (e.g., 0.1–2.4 keV and 0.3–2 keV) using $f_\nu \propto \nu^{-\alpha_x}$ ($\alpha_x = \Gamma_x - 1$, Γ_x is the power-law photon index). The typical values of Γ_x , taken from D. Donato et al. (2005), are adopted as 1.59, 1.92, and 2.24 for FSRQs, LBLs+IBLs, and HBLs, respectively. The X-ray fluxes of our blazars are mainly obtained from NED, with a few taken from the literatures. The broad-band spectral indices (α_{ro} , α_{ox} , and α_{rx}) are calculated using the equation as the follow (J. E. Ledden & S. L. O'Dell 1985):

$$\alpha_{ij} = -\log(f_i/f_j)/\log(\nu_i/\nu_j) \quad (1)$$

where f_ν and ν correspond to the flux and frequency, respectively.

The strength of jet is the most intuitive indicator of jet properties, and can be quantified as $P_{\text{jet}}^{\text{in}}$ (e.g., E. T. Meyer et al. 2011; M. Keenan et al. 2021). Usually, $P_{\text{jet}}^{\text{in}}$ is represented by the low frequency radio luminosity (L_r^{low}), as it is only marginally affected by Doppler-boosted compact components (Z. Wu et al. 2007; M. Keenan et al. 2021). The formula is as follow (e.g., K. W. Cavagnolo et al. 2010)

$$\log P_{\text{jet}}^{\text{in}} = 0.64(\log L_r^{\text{low}} - 40) + 43.54 \quad (2)$$

⁴ <https://ned.ipac.caltech.edu/>

⁵ <https://simbad.u-strasbg.fr/simbad/>

In this work, we adopt the 340 MHz radio fluxes ($f_{340\text{MHz}}$) to calculate $P_{\text{jet}}^{\text{in}}$, which are taken from CDS. For blazars without available $f_{340\text{MHz}}$, the other approximate wave bands (e.g., 150 MHz, 330 MHz, 352 MHz, 365 MHz, and 843 MHz) are extrapolated to $f_{340\text{MHz}}$ using $f_{\nu} \propto \nu^{-\alpha_r}$ ($\alpha_r = 0.7$, e.g., [M. Keenan et al. 2021](#)).

The Doppler beaming effect, quantified by δ_{ν} , is often considered in blazar physics. However, δ_{ν} has generally been estimated in the radio and γ -ray bands (e.g., [G. Ghisellini et al. 1993](#); [T. Hovatta et al. 2009](#); [Z. Wu et al. 2007, 2014](#); [J.-H. Fan et al. 2014](#); [I. Liodakis et al. 2017, 2018](#); [Z. Pei et al. 2022](#)), whereas estimates are generally unavailable in the optical and X-ray bands. The δ_r of our blazars are taken from [T. Hovatta et al. \(2009\)](#) (22 GHz & 37 GHz), [Z. Wu et al. \(2014\)](#) (5 GHz), [I. Liodakis et al. \(2017\)](#) (see their table 1), [I. Liodakis et al. \(2018\)](#) (15 GHz), and [X.-H. Ye & J.-H. Fan \(2021\)](#) (5 GHz). In theory, δ_{ν} is determined by the velocity (β) of plasma flows and the view angle (θ) to our line of sight: $\delta_{\nu} = [\Gamma(1 - \beta\cos\theta)]^{-1}$, where Γ is the bulk Lorentz factor ($\Gamma = 1/\sqrt{1 - \beta^2}$). Considering the possible acceleration and deceleration of blazar jets, δ_{ν} may vary among different emission bands. Because we adopt the 5 GHz core radio emissions in this paper, δ_{ν} at 5 GHz band ($\delta_{5\text{GHz}}$) is adopted. Conservatively, we extrapolate δ_{ν} at other bands to $\delta_{5\text{GHz}}$ using the empirical formula of accelerated and decelerated jet model (see [J. H. Fan et al. 1994](#); [Q.-C. Long et al. 2026](#)):

$$\begin{cases} \delta_{\nu} = \delta_o^{1 + \frac{1}{8} \log \frac{\nu_o}{\nu}} & (\alpha_{\text{rx}} > 0.75) \\ \delta_{5\text{GHz}} = 10^{\log(\frac{\delta_{\nu}}{\nu^{0.07}}) + 0.682} & (\alpha_{\text{rx}} < 0.75) \end{cases} \quad (3)$$

In the first item, δ_o is the optical Doppler factor at the typical optical band ($\nu_o = 10^{14}$ Hz, see [J. H. Fan et al. 1994](#)). This item is appropriate only to the accelerated jet model for strong-jet sources (FSRQs+LBLs [J. H. Fan et al. 1994](#); [I. Liodakis et al. 2018](#)). The second item corresponds to the decelerated jet model of HBLs ([Q.-C. Long et al. 2026](#)). For IBLs, they may be the transitional objects between LBLs and HBLs ([E. Nieppola et al. 2006](#); [A. A. Abdo et al. 2010](#); [E. T. Meyer et al. 2011](#)). Therefore, for IBLs with $\alpha_{\text{rx}} > 0.75$ (IBL $_{>0.75}$), we apply the first item of Eq (3), otherwise (i.e., IBL $_{<0.75}$), we apply the second item. Note that we consider Eq (3) to be applicable for extrapolations between radio bands. However, extrapolation across widely separated bands (e.g., $\delta_r \rightarrow \delta_o$ or δ_x) would introduce substantially larger uncertainties and was therefore avoided. If the sources have multiple $\delta_{5\text{GHz}}$, we will take the average values $\langle \delta_{5\text{GHz}} \rangle$ of them. Note that values of $\delta_r < 1$ are excluded, as they would imply the observational emissions (f_{ν}^{ob}) weaker than the intrinsic ones (f_{ν}^{in} ; see Eq (4), corresponding to Doppler de-boosting. Such a scenario is generally considered unlikely for blazars and is more likely to arise from the large uncertainties associated with estimates of δ_r . Excluding $\delta_r < 1$ measurements does not affect our results. This is because, in the initial cross-matched sample, only a very small fraction of the sources have reported $\delta_r < 1$ values. Among these sources, those with available broad-band data are still included in our sample; only the corresponding $\delta_r < 1$ values are excluded from the analysis.

L_{disk} is difficult to measure for blazars because their entire electromagnetic spectrum is dominated by non-thermal radiation. For blazars, L_{disk} is usually derived from the luminosity of broad line region (BLR), L_{BLR} , by assuming $L_{\text{disk}} = 10L_{\text{BLR}}$ (e.g., [G. Ghisellini et al. 2014](#); [V. S. Paliya et al. 2021](#)). However, most BL Lacs have no emission lines, making the absence of their L_{disk} . To obtain the more reliable λ_{Edd} , we made our best effort to search L_{BLR} and M_{BH} of our blazars in the literatures (the references see Table 1). However, we find that there are 127/136 FSRQs, 34/64 LBLs, 8/39 IBLs, and 42/105 HBLs that have both available L_{disk} and M_{BH} in our sample (the composition of our sample is summarized in Table 3). Excluding sources lacking both L_{disk} and M_{BH} may improve reader confidence, yet it would drastically reduce the BL Lac sample size, which would severely impact our statistical results (the more detailed discussions see Appendix A). Therefore, for blazars without available L_{disk} , we try to investigate a correlation between L_{disk} and other luminosities to estimate their L_{disk} . Fortunately, we find a strong correlation between L_{disk} and L_{x}^{ob} for FSRQs+LBLs+IBL $_{>0.75}$ and HBLs+IBL $_{<0.75}$, respectively (see Eq (A1) and Eq (A2)). Therefore, for blazars without available L_{disk} , we use Eq (A1) and Eq (A2) to estimate their L_{disk} .

The M_{BH} of our blazars are taken from the existing literatures (see Table 1). The traditional virial BH mass is also known as the dynamical mass, which adopts an empirical relationship between the size of BLR and the continuum luminosity, as well as the measured broad-line width, assuming that the BLR clouds gravitationally bound to the central BH ([M. Vestergaard & B. M. Peterson 2006](#)). This method is usually applied to estimate M_{BH} for FSRQs and some LBLs that show well-detected broad lines (e.g., [M. S. Shaw et al. 2012](#)). For HBLs, IBLs, and some LBLs, they have no or weak emission line, so their M_{BH} are usually estimated from the properties of their host galaxies, e.g., $M_{\text{BH}} - \sigma$ and $M_{\text{BH}} - L_{\text{bulge}}$ relations (where σ and L_{bulge} are the stellar velocity dispersion and the bulge luminosity of the host galaxies, respectively). For blazars without available M_{BH} , we used the fundamental plane (FP) of black hole

activity to estimate their M_{BH} (e.g., K. Gültekin et al. 2009; L. G. C. Bariuan et al. 2022). The Appendix A provides the relevant knowledge about FP. Similarly, we divide our blazars into FSRQs+LBLs+IBL $_{>0.75}$ and HBLs+IBL $_{<0.75}$, the best fit of their FPs are Eq (A4) and Eq (A5), respectively. Therefore, for blazars without available M_{BH} , we use Eq (A4) and Eq (A5) to estimate their M_{BH} . Noting, the means above do not have a significant effect on our results in the next (see Appendix A).

The basic parameters of first sources of FSRQs, LBLs, IBLs, and HBLs are given in the representative Table 1. The complete table is available online in machine-readable format. Figure 1 shows the distributions and comparisons of properties for our blazars.

Table 1. The Properties of Blazars

IAU Name	z	$\log \nu_p^{\text{syn}}$	Class	$\log L_x^{\text{ob}}$	$\log L_o^{\text{ob}}$	$\log L_r^{\text{ob}}$	$\log P_{\text{jet}}^{\text{in}}$	$\log L_{\text{disk}}$	$\log M_{\text{BH}}$	Refs.	α_{ox}	α_{ro}	α_{rx}	$\delta_{5\text{GHz}}$	$\langle \delta_{5\text{GHz}} \rangle$	λ_{Edd}
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
0007+205	0.600	12.56	FSRQ	44.48	44.70	42.83	44.90	45.35	7.86	1/2/1/2/3/3	1.081	0.630	0.785	.../6.47/.../.../...	6.47	-0.624
...	...	...	...	...	...	...	...	...	...	.../.../.../.../...	...	...	...	.../.../.../.../...	...	...
0003-066	0.347	12.92	LBL	44.05	44.91	43.58	45.16	44.52	8.93	1/2/1/2/3/3	1.323	0.737	0.939	5.40/7.53/.../5.97/5.58	6.12	-2.524
...	...	...	...	...	...	...	...	...	...	.../.../.../.../...	...	...	...	.../.../.../.../...	...	...
0048-09	0.634	14.12	IBL	45.18	45.97	43.89	45.21	...	8.85	1/2/9/2/.../3	1.296	0.587	0.832	8.46/22.84/12.10/4.58/10.88	11.77	-0.948
...	...	...	...	...	...	...	...	...	...	.../.../.../.../...	...	...	...	.../.../.../.../...	...	...
0032+595	0.086	17.05	HBL	44.40	43.21	40.61	43.75	...	7.26	1/2/9/2/.../29	0.549	0.484	0.506	3.19/.../.../2.33/...	2.76	-1.565
...	...	...	...	...	...	...	...	...	...	.../.../.../.../...	...	...	...	.../.../.../.../...	...	...

NOTE—Col (1): IAU Name. Col (2): Redshift. Col (3): logarithm of ν_p^{syn} for blazars (Hz). Col (4): the classes of blazars. Col (5): logarithm of the observational 1 keV X-ray luminosity (L_x^{ob} in units of erg s^{-1}). Col (6): logarithm of the observational optical luminosity at 5.4143×10^{14} Hz V-band (L_o^{ob} in units of erg s^{-1}). Col (7): logarithm of the observational 5 GHz core radio luminosity (L_r^{ob} in units of erg s^{-1}). Col (8): logarithm of the intrinsic jet power ($P_{\text{jet}}^{\text{in}}$ in units of erg s^{-1}). Col (9): logarithm of the accretion disk luminosity (L_{disk} in units of erg s^{-1}). Col (10): logarithm of the dynamical BH mass (M_{BH} in units of $M_{\odot}$). Col (11): references of the X-ray flux, optical flux, 5 GHz core radio flux, low frequency radio flux, accretion disk luminosity, and BH mass. Col (12), Col (13), and Col (14) are the broad-band spectral indices from optical 5.4143×10^{14} Hz V-band to 1 keV X-ray band (α_{ox}), from 5 GHz radio band to optical 5.4143×10^{14} Hz V-band (α_{ro}), and from 5 GHz radio band to 1 keV X-ray band (α_{rx}), respectively. Col (15): the 5 GHz radio Doppler factors refer from X.-H. Ye & J.-H. Fan (2021)/I. Lioudakis et al. (2018)/I. Lioudakis et al. (2017)/Z. Wu et al. (2014)/T. Hovatta et al. (2009) (Note, δ_r at other radio band have been extrapolated to $\delta_{5\text{GHz}}$). Col (16): the average 5 GHz Doppler factor $\langle \delta_{5\text{GHz}} \rangle$. Col (17): the Eddington-ratio λ_{Edd} . References: (1) NED; (2) CDS; (3) V. S. Paliya et al. (2021); (4) G. Chen et al. (2024); (5) M. Sikora et al. (2007); (6) B. Chai et al. (2012); (7) M. Zhou & X.-W. Cao (2009); (8) R. D. Saxton et al. (2008); (9) X.-H. Ye & J.-H. Fan (2021); (10) Y. Liu et al. (2006); (11) D. Xiong et al. (2015a); (12) R. Wang et al. (2006); (13) T. Dwelly et al. (2017); (14) Y. Shen et al. (2011); (15) N. Kurinsky et al. (2013); (16) Z. Yuan & J. Wang (2012); (17) M. Gu et al. (2009); (18) F. Mantovani et al. (2015); (19) Q.-C. Long et al. (2025); (20) H. L. Marshall et al. (2018); (21) T. Sbarato et al. (2012); (22) Z. Yuan et al. (2018); (23) Y. Y. Kovalev et al. (2020); (24) J.-H. Woo & C. M. Urry (2002); (25) E. Pian et al. (2005); (26) M_{R} from K. Nilsson et al. (2003), with M_{BH} estimated via the $M_{\text{BH}} - L_{\text{bulge}}$ relation of A. W. Graham (2007); (27) W. Brinkmann et al. (1997); (28) G. Ghisellini et al. (2011); (29) Z.-Z. Wu et al. (2009); (30) Z. Wu et al. (2014); (31) Z.-H. Fan & X. Cao (2004); (32) M_{R} from M. O'Dowd & C. M. Urry (2005), with M_{BH} estimated via the $M_{\text{BH}} - L_{\text{bulge}}$ relation of A. W. Graham (2007); (33) R. M. Plotkin et al. (2011); (34) R. Falomo et al. (2003); (35) Q.-C. Long et al. (2026); (36) J.-M. Wang et al. (2002).

Table 2. The Average Properties for Blazars

Class	$\langle \log \nu_p^{\text{syn}} \rangle$	$\langle z \rangle$	$\langle \alpha_{\text{ox}} \rangle$	$\langle \alpha_{\text{ro}} \rangle$	$\langle \alpha_{\text{rx}} \rangle$	$\langle \log L_x^{\text{ob}} \rangle$	$\langle \log L_o^{\text{ob}} \rangle$	$\langle \log L_r^{\text{ob}} \rangle$	$\langle \log P_{\text{jet}}^{\text{in}} \rangle$	$\langle \delta_{5\text{GHz}} \rangle$	$\langle \lambda_{\text{Edd}} \rangle$
FSRQ	12.89	1.094	1.248	0.709	0.895	45.07	45.73	44.27	45.57	15.90	-0.959
LBL	13.21	0.515	1.338	0.646	0.884	44.05	44.95	43.16	44.86	10.09	-2.089
IBL	14.51	0.380	1.356	0.445	0.759	43.92	44.87	42.07	44.33	5.31	-2.695
HBL	16.38	0.250	1.060	0.404	0.630	44.09	44.24	41.24	43.89	2.84	-3.490

3. RESULTS

3.1. The Distribution and Comparison of Properties among Blazars

The underlying physics of observation properties for blazars can be obtained from the fundamental physics parameters, which can reveal the similarities and differences among them. Table 2 provides the average properties of blazars. Figure 1 shows the distributions and comparisons of parameters for our blazars.

Table 2 shows that all average properties, but α_{ox} and $\log L_x^{\text{ob}}$, are negatively correlated with ν_p^{syn} for blazar sequence. Figure 1-(a) shows a rough anti-correlation between redshift and $\log \nu_p^{\text{syn}}$ for blazars, with a Pearson correlation coefficient of $R = -0.547$ and $P = 3.01 \times 10^{-28}$. This finding is consistent with G. Fossati et al. (1998), suggesting that FSRQs are more readily detected at higher z .

Figure 1-(b) shows a rough anti-correlation between $\langle \delta_{5\text{GHz}} \rangle$ and $\log \nu_p^{\text{syn}}$ for blazar sequence, with a Pearson correlation coefficient of $R = -0.621$ and $P = 4.65 \times 10^{-38}$. This finding is consistent with Z. Wu et al. (2007), E. Nieppola et al. (2008), W. X. Yang et al. (2022), and Q.-C. Long et al. (2025). Because $\delta \approx \Gamma$ (G. Ghisellini et al. 2014). Combining the VLBA observations discussed in Section 1, the observed $\delta_r - \nu_p^{\text{syn}}$ anti-correlation can be interpreted in terms of an accelerating/decelerating jet scenario across different blazar subclasses. At least for HBLs, their jets may have already decelerated to sub-relativistic speeds and become decollimated by the time they propagate to the radio region, whereas the jets in FSRQs and most LBLs still maintain highly relativistic speeds on radio scales. Note, there are 15 FSRQs that have $\langle \delta_{5\text{GHz}} \rangle > 30$, with a maximum value of $\langle \delta_{5\text{GHz}} \rangle = 105.97$ for 0446+112. To mitigate the significant disparity and obtain a more continuous distribution of $\langle \delta_{5\text{GHz}} \rangle$ in Figure 1-(b), Figure 2, and Figure 3, we assigned a cap value of $\langle \delta_{5\text{GHz}} \rangle = 30$ to these extreme outliers. This measure does not have a significant effect on our results.

Figure 1-(c) shows an anti-correlation between $\log P_{\text{jet}}^{\text{in}}$ and $\log \nu_p^{\text{syn}}$ for blazar sequence, with a Pearson correlation coefficient of $R = -0.757$ and $P = 3.31 \times 10^{-65}$. This suggests different jet strength among blazars, the blazars with a lower ν_p^{syn} have a more powerful jet.

Figure 1-(d) shows a V-type evolution in $\log L_x^{\text{ob}} - \log \nu_p^{\text{syn}}$ relation for blazar sequence, which agrees with E. Nieppola et al. (2006), J. H. Fan et al. (2016), and J. H. Yang et al. (2022). The break occurs at about $\nu_p^{\text{syn}} \sim 10^{14.5}$ Hz, where IBLs seem to serve as a bridge between FSRQs+LBLs and HBLs.

Figure 1-(g) shows a $\wedge$ -type evolution in $\alpha_{\text{ox}} - \log \nu_p^{\text{syn}}$ relation for blazar sequence (here, IBLs also seem to serve as a bridge between FSRQs+LBLs and HBLs), which is opposite to the $\log L_x^{\text{ob}} - \log \nu_p^{\text{syn}}$ relation. This finding is consistent with J. H. Fan et al. (2016) and J. H. Yang et al. (2022).

Figure 1-(h) shows an anti-correlation between α_{ro} and $\log \nu_p^{\text{syn}}$ in the sequence of FSRQs→LBLs→IBLs, with a Pearson correlation coefficient of $R = -0.714$ and $P = 1.39 \times 10^{-38}$. HBLs show an almost horizontal line in $\alpha_{\text{ro}} - \log \nu_p^{\text{syn}}$ relation. This finding roughly agrees with G. Fossati et al. (1998), J. H. Fan et al. (2016), and J. H. Yang et al. (2022).

Figure 1-(i) shows an anti-correlation between α_{rx} and $\log \nu_p^{\text{syn}}$ for blazar sequence, with a Pearson correlation coefficient of $R = -0.875$ and $P = 7.26 \times 10^{-110}$. This finding is consistent with G. Fossati et al. (1998).

Figure 1-(j) shows an anti-correlation between $\log L_{\text{disk}}$ and $\log \nu_p^{\text{syn}}$ for blazars with available L_{disk} , with a Pearson correlation coefficient of $R = -0.438$ and $P = 9.63 \times 10^{-23}$.

Figure 1-(k) shows an almost horizontal line in $\log M_{\text{BH}} - \log \nu_p^{\text{syn}}$ relation for blazars with available M_{BH} , with a Pearson correlation coefficient of $R = 0.094$ and $P = 0.115$.

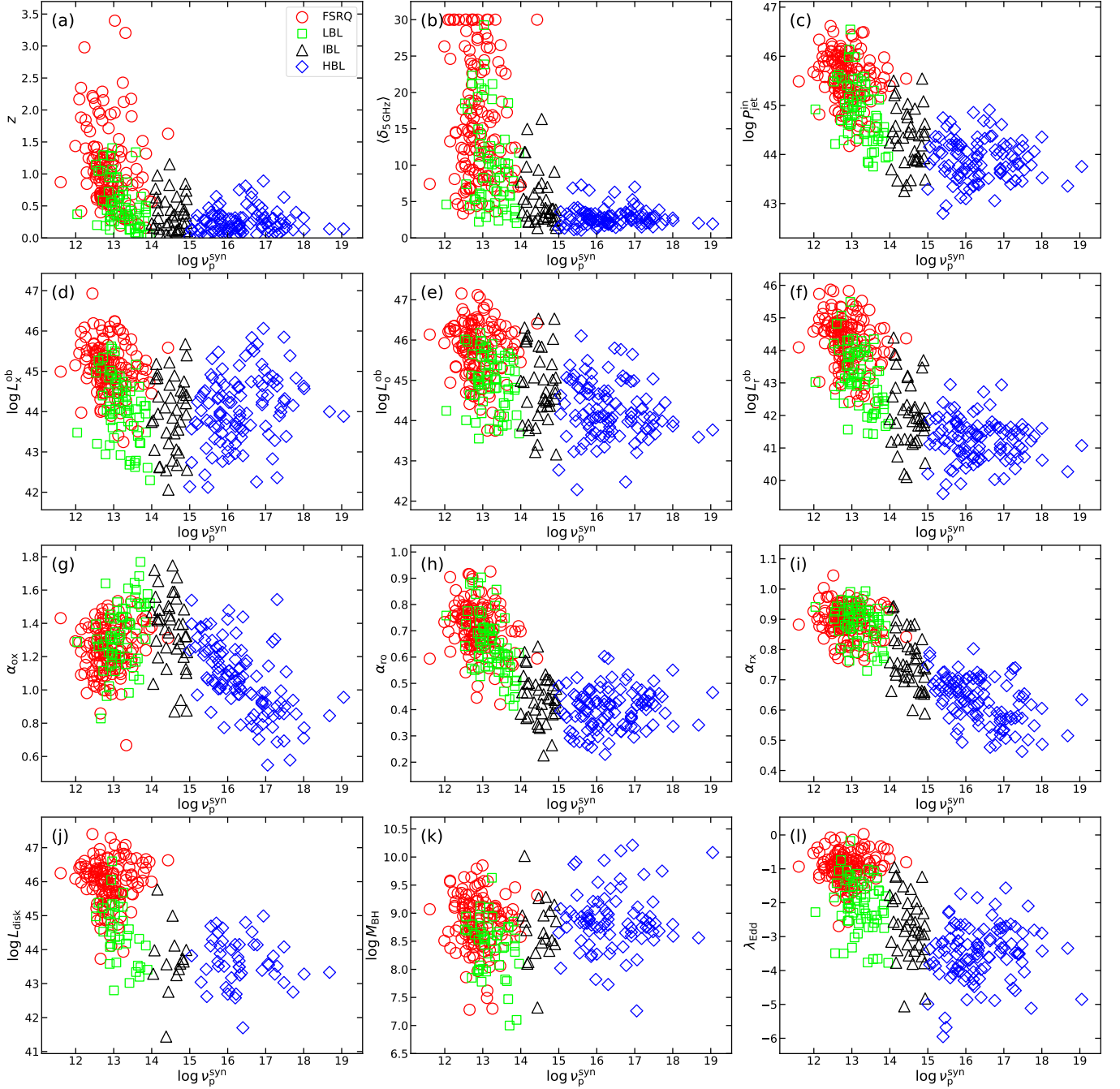


Figure 1. The relation between ν_p^{syn} and properties for blazar sequence. The red circles, lime squares, black triangles, and blue diamonds are FSRQs, LBLs, IBLs, and HBLs, respectively. (a) z vs. $\log \nu_p^{\text{syn}}$ diagram. (b) $\langle \delta_{5\text{GHz}} \rangle$ vs. $\log \nu_p^{\text{syn}}$ diagram. (c) $\log P_{\text{jet}}^{\text{in}}$ vs. $\log \nu_p^{\text{syn}}$ diagram. (d) $\log L_x^{\text{ob}}$ vs. $\log \nu_p^{\text{syn}}$ diagram. (e) $\log L_o^{\text{ob}}$ vs. $\log \nu_p^{\text{syn}}$ diagram. (f) $\log L_r^{\text{ob}}$ vs. $\log \nu_p^{\text{syn}}$ diagram. (g) α_{ox} vs. $\log \nu_p^{\text{syn}}$ diagram. (h) α_{ro} vs. $\log \nu_p^{\text{syn}}$ diagram. (i) α_{rx} vs. $\log \nu_p^{\text{syn}}$ diagram. (j) $\log L_{\text{disk}}$ vs. $\log \nu_p^{\text{syn}}$ diagram. (k) $\log M_{\text{BH}}$ vs. $\log \nu_p^{\text{syn}}$ diagram. (l) λ_{Edd} vs. $\log \nu_p^{\text{syn}}$ diagram.

Figure 1-(l) shows an anti-correlation between λ_{Edd} and $\log \nu_p^{\text{syn}}$ for blazar sequence, with a Pearson correlation coefficient of $R = -0.736$ and $P = 5.06 \times 10^{-60}$. This finding agrees with V. S. Paliya et al. (2021), suggesting different accretion regimes among blazars. The blazars with a lower ν_p^{syn} have higher accretion rates.

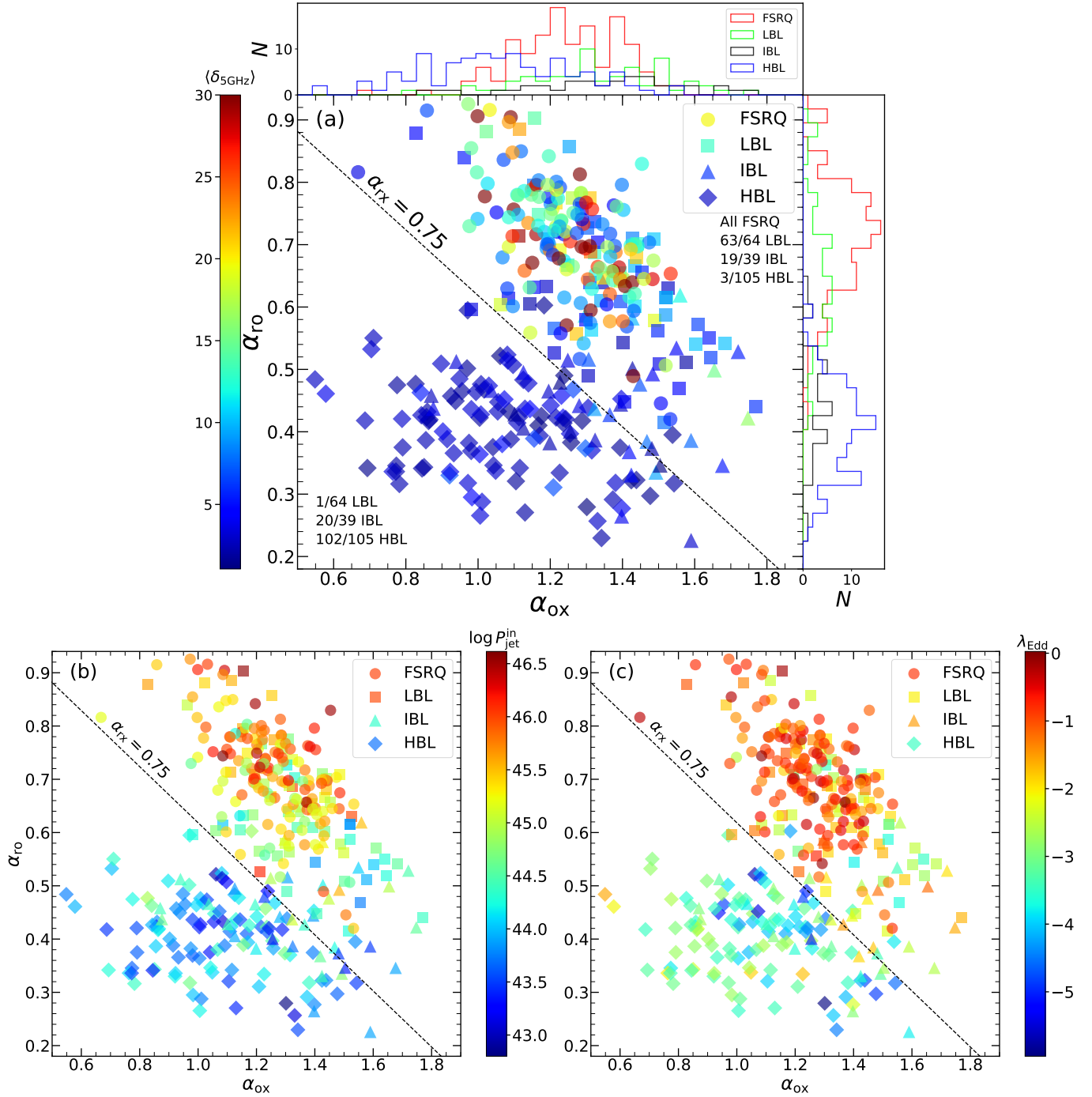


Figure 2. The α_{ro} vs. α_{ox} diagram. The circles, squares, triangles, and diamonds are FSRQs, LBLs, IBLs, and HBLs, respectively. α_{rx} can be regarded as the intercept; the sources that occupy the upper right region have $\alpha_{\text{rx}} > 0.75$, on the contrary, the sources that occupy the lower left corner have $\alpha_{\text{rx}} < 0.75$. (a), (b), and (c) show Doppler beaming effect at 5 GHz $\langle \delta_{5\text{GHz}} \rangle$, intrinsic jet power $\log P_{\text{jet}}^{\text{in}}$, and accretion rates λ_{Edd} , respectively, with color scales representing each quantity. The histograms on the axes in plane-(a) show the distribution of this physical quantities (α_{ox} and α_{ro}). The red, lime, black, and blue box are FSRQs, LBLs, IBLs, and HBLs, respectively.

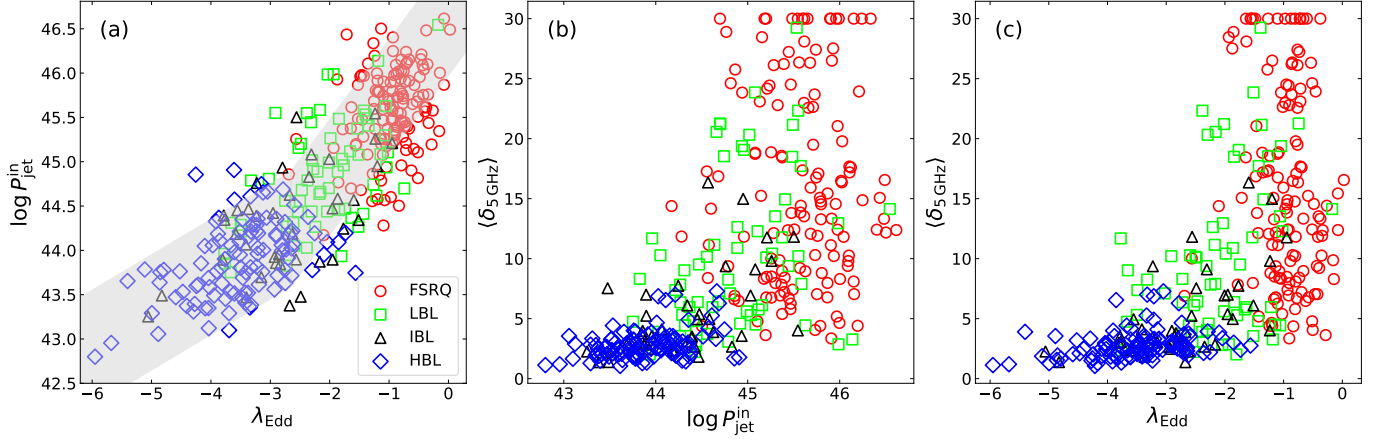


Figure 3. The comparison between intrinsic jet power ($\log P_{\text{jet}}^{\text{in}}$), Eddington-radio (λ_{Edd}), and 5 GHz Doppler factor ($\langle \delta_{5\text{GHz}} \rangle$) for our sample. (a) $\log P_{\text{jet}}^{\text{in}}$ vs. λ_{Edd} diagram; (b) $\langle \delta_{5\text{GHz}} \rangle$ vs. $\log P_{\text{jet}}^{\text{in}}$ diagram; (c) $\langle \delta_{5\text{GHz}} \rangle$ vs. λ_{Edd} diagram. The notations are the same as that of Figure 1.

3.2. The Distribution of Doppler Beaming Effect, Jet Power, and Accretion Rates on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ Plane

To achieve our aims (see section 1), we put our blazars into the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane and confer them with $\langle \delta_{5\text{GHz}} \rangle$, $P_{\text{jet}}^{\text{in}}$, and λ_{Edd} values in the scale of color, our results are shown in Figure 2.

It is clear that FSRQs+LBLs→IBLs→HBLs follow a Δ -shaped evolutionary trajectory, which is consistent with previous studies (e.g., P. Padovani & P. Giommi 1995; D. Donato et al. 2001; A. A. Abdo et al. 2010; J. H. Fan et al. 2016; J. H. Yang et al. 2022). In the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane, α_{rx} can be regarded as the intercept (see J. E. Ledden & S. L. O'Dell 1985). In other words, α_{rx} increases toward the upper-right of the plane. Our results show that all FSRQs reside in the upper right region (URR) with $\alpha_{\text{rx}} > 0.75$, there are 63 out of 64 (63/64) LBLs that occupy the URR, only 1/64 LBL occupies the lower left region (LLR) with $\alpha_{\text{rx}} < 0.75$. In addition, the region inhabited by FSRQs significantly overlaps with that of LBLs. IBLs are almost equally distributed between URR (19/39) and LLR (20/39). There are 102/105 HBLs that occupy LLR, only 3/105 HBLs occupy URR. IBLs seem to be the transitional objects between FSRQs+LBLs and HBLs. The statistics above are presented in Figure 2-(a).

The histograms on the axes of Figure 2-(a) show the distribution of α_{ox} and α_{ro} . On average, the distribution of α_{ox} of FSRQs+LBLs is only slightly larger than that of HBLs. However, the distributions of α_{ro} and α_{rx} of FSRQs+LBLs are significantly larger than those of HBLs (see also Figure 1 (g), (h), and (i)).

From Figure 2-(a), we find a nearly continuous distribution of $\langle \delta_{5\text{GHz}} \rangle$ for blazar sequence, in which $\langle \delta_{5\text{GHz}} \rangle$ decreases toward the Δ -shaped track. $P_{\text{jet}}^{\text{in}}$ and λ_{Edd} also decrease toward this track (see Figure 2 (b) and (c)). It is clear that there are the positive correlations among $\langle \delta_{5\text{GHz}} \rangle$, $P_{\text{jet}}^{\text{in}}$ and λ_{Edd} . For a more intuitive illustration, we present the plots in Figure 3. Figure 3-(a) shows a break at $\lambda_{\text{Edd}} \sim -2.651$ in the $\log P_{\text{jet}}^{\text{in}} - \lambda_{\text{Edd}}$ relation, which is estimated by visual inspection. This critical value is consistent with the theoretical transition value between the radiatively efficient and radiatively inefficient accretion regimes (e.g., G. Ghisellini & F. Tavecchio 2008; F. Yuan & R. Narayan 2014).

4. DISCUSSION

Before proceeding with further discussion, we make the following assumptions, which have also been adopted in many previous works (e.g., M. Chiaberge et al. 2000; C. M. Raiteri et al. 2017). We assume that the radio, optical, X-ray emissions are dominated by jets, and they come from different regions along the jet, as shown in Figure 4 (see also C. M. Raiteri et al. 2017).

4.1. The Physical Properties Behind the Distribution of Blazars on $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane

Figure 2 implies that the distribution of blazars on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane is regulated by the accretion rates and the Doppler beaming effect. Next, we will conduct a further analysis from a theoretical perspective. In the beaming model, f_{ν}^{ob} is strongly boosted from f_{ν}^{in} , an empirical equation is given as:

$$f_{\nu}^{\text{ob}} = \delta_{\nu}^q f_{\nu}^{\text{in}} \quad (4)$$

where $q = 2 + \alpha_\nu$ for the continuous jet and $q = 3 + \alpha_\nu$ for the spherical jet (P. A. G. Scheuer & A. C. S. Readhead 1979), α_ν is the spectral index at ν band. δ_ν is Doppler factor at ν band. Combining Eq (1) with Eq (4), we have:

$$\alpha_{ij} = -\log[(f_i^{\text{in}} \delta_i^{q_i}) / (f_j^{\text{in}} \delta_j^{q_j})] / \log(\nu_i / \nu_j) \quad (5)$$

Taking $\nu_r = 5 \times 10^9$ Hz, $\nu_o = 5.4143 \times 10^{14}$ Hz, and $\nu_x = 2.41799 \times 10^{17}$ Hz into Eq (5), we can obtain:

$$\begin{cases} \alpha_{ro} = 0.199 \log\left(\frac{f_r^{\text{in}} \delta_r^{q_r}}{f_o^{\text{in}} \delta_o^{q_o}}\right) \\ \alpha_{ox} = 0.377 \log\left(\frac{f_o^{\text{in}} \delta_o^{q_o}}{f_x^{\text{in}} \delta_x^{q_x}}\right) \\ \alpha_{rx} = 0.130 \log\left(\frac{f_r^{\text{in}} \delta_r^{q_r}}{f_x^{\text{in}} \delta_x^{q_x}}\right) \end{cases} \quad (6)$$

It is clear that α_{ij} is determined by the ratio of $f_i^{\text{in}}/f_j^{\text{in}}$ and $\delta_i^{q_i}/\delta_j^{q_j}$, where $\delta_i^{q_i}/\delta_j^{q_j}$ is determined by δ_i/δ_j , jet shape ($q = 2(3) + \alpha_\nu$), and α_ν .

In theoretical model, the relativistic jets are produced from the innermost regions of the accretion disk, so the jet variables should depend on two fundamental parameters that determine the conditions in the inner accretion disc, namely dimensionless accretion rate ($\dot{m} = \dot{M}/\dot{M}_{\text{Edd}}$, $\dot{M}_{\text{Edd}}$ is the Eddington accretion rate) and M_{BH} (S. Heinz & R. A. Sunyaev 2003):

$$f_\nu^{\text{in}} \propto M_{\text{BH}}^{\xi_M} \dot{m}^{\xi_{\dot{m}}} \propto \dot{M}^{\xi_\nu} \quad (7)$$

this gives rise to the scaling relation:

$$f_i^{\text{in}} / f_j^{\text{in}} \propto \dot{M}^{\xi_{ij}} \quad (8)$$

where ξ_{ij} is regulated by accretion mode (see S. Heinz & R. A. Sunyaev 2003; A. Merloni et al. 2003; F. Yuan & W. Cui 2005; Q.-C. Long et al. 2025). Therefore, our observational results are consistent with the theoretical expectation.

However, $f_i^{\text{in}}/f_j^{\text{in}}$ of blazars are very difficult to obtain due to the fact that δ_ν is an unobservable quantity. Therefore, a coming question is “Is α_{ij} more sensitive to $f_i^{\text{in}}/f_j^{\text{in}}$ or to $\delta_i^{q_i}/\delta_j^{q_j}$?” Table 2 and Figure 2 show that the average value and distribution of α_{ox} of FSRQs+IBLs are only slightly larger than those of HBLs, even if there is a significant difference in their λ_{Edd} (see Figure 1-(l)). But when the radio emissions are taken into account, the remarkable differences are found in the distribution of α_{ro} and α_{rx} , which suggests that δ_r plays a larger role than the accretion rates in influencing α_{ij} . This implies that the difference in jet properties between FSRQs+LBLs and HBLs may occur in the region from optical to radio (IBLs are considered transitional sources). The jets of HBLs may begin to decelerate in the optical region and exhibit sub-relativistic speeds by the time they propagate to the radio region. The jets of FSRQs+LBLs still maintain relativistic speeds down to the radio region. This analysis agrees with the results that are derived from Figure 1-(b) (see Section 3). Therefore, δ_r/δ_x and δ_r/δ_o of FSRQs+LBLs should be larger than those of HBLs. The similar properties were found in radio galaxies, G. Giovannini et al. (2001) found that the jets of FR Is and FR IIs are both highly relativistic in the parsec (pc) scale, but there are differences on kiloparsec (kpc) scale. The jets of FR Is may be strongly decelerated within several hundred pc from the core, and often exhibit the sub-relativistic speeds and decollimated characteristic as they propagate out to kpc scales (A. Bhattacharjee et al. 2024). while the jets of FR IIs appear well-collimated characteristic and relativistic speeds up to ~ 100 kpc (B. Boccardi et al. 2021; J. Seo et al. 2021).

Continuous jet ($q = 2 + \alpha_\nu$) vs. spherical jet ($q = 3 + \alpha_\nu$). Historically, the continuous jet refers to the steady jet, while the spherical jet is considered as a sum of discrete features known as blobs or knots, i.e., the so-called intermittent jet (e.g., M.-Á. Aloy et al. 2003; R. P. Fender et al. 2004; J. Ferreira et al. 2006), where the compact plasma blobs are inlaid in the collimated outflows (M.-Á. Aloy et al. 2003; J. Ferreira et al. 2006; T. Vuillaume et al. 2018). BH X-ray binaries (BH-XRBs) are the typical laboratory to study the continuous jet and intermittent jet (e.g., R. P. Fender et al. 2004; J. Ferreira et al. 2006; R. Fender & T. Belloni 2012). Continuous jets occur in the low/hard state of BH-XRBs, where the accretion disk is usually described by ADAF. With the increase of accretion rate, BH-XRBs enter the luminous intermediate state, in which the thin disk almost extending down to the innermost stable circular orbit, and the intermittent jets emerge (R. P. Fender et al. 2004; J. Ferreira et al. 2006; R. Fender & T. Belloni 2012; M. C. Begelman & P. J. Armitage 2014). The intermittent jet has a larger bulk Lorentz factor (see Figure 7 in R. P. Fender et al. 2004). The ultra-relativistic blobs, which are produced when the adequate conditions are fulfilled (J.

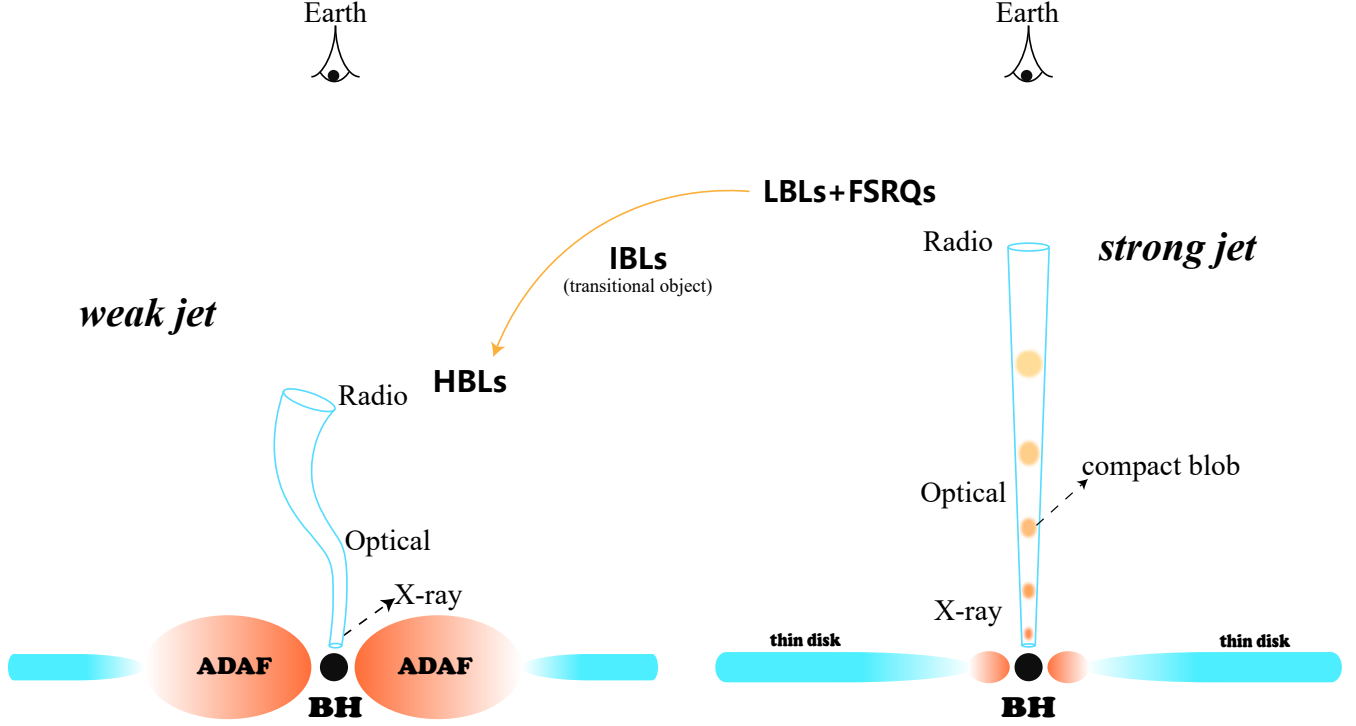


Figure 4. The cartoon picture of blazar sequence. This accretion disk mode refers from the gradual evolution of accretion disk in BH-XRB (e.g., R. P. Fender et al. 2004; J. Ferreira et al. 2006; R. Fender & T. Belloni 2012; M. C. Begelman & P. J. Armitage 2014). Noting, the accretion disk in the left is dominated by ADAF instead of the hybrid mode of ADAF+SSD. The main difference of jets between FSRQs+LBLs and HBLs occur in the region from optical to radio. In addition, the jet base of FSRQs+LBLs has a capacity to emit the compact blobs, these blobs moving ultra-relativistically in the channel of outflows.

Ferreira et al. 2006), have been used to explain the highly relativistic phenomena such as super-luminal motions in quasars (G. Pelletier & J. Roland 1989; G. Pelletier & H. Sol 1992; M.-Á. Aloy et al. 2003).

The blobs are formed from the electron-positron ($e^+ - e^-$) pairs (G. Henri & G. Pelletier 1991; J. Ferreira et al. 2006). Theoretically, γ -ray photons with energies near threshold ($\hbar\omega \gtrsim 2m_e c^2$) can annihilate with themselves to produce $e^+ - e^-$ pairs in superstrong magnetic fields ($\phi_B \gtrsim m_e^2 c^3 / e\hbar$, see J. K. Daugherty & A. K. Harding 1983), where the squared magnetic fields are proportional to the accretion rates ($\phi_B^2 \propto \dot{M}$, e.g., S. Heinz & R. A. Sunyaev 2003; G. Ghisellini et al. 2014; M. Zamaninasab et al. 2014). Continuous generation of $e^+ - e^-$ pairs accumulates to form $e^+ - e^-$ plasma clouds, ultimately giving rise to blobs (R. Svensson 1987; G. Henri & G. Pelletier 1991). Then the blobs are accelerated by the Compton rocket effect (S. L. O’Dell 1981; T. Vuillaume et al. 2018). Therefore, the ultra-relativistic blobs are usually produced in the jets of high accretion BH. Hence, jets of FSRQs+LBLs should be intermittent jet ($q = 3 + \alpha_\nu$) in which the compact blobs moving ultra-relativistically in the channel of outflows, while the jets of HBLs should seem to be the continuous jet ($q = 2 + \alpha_\nu$). Indeed, there were many VLBA studies of pc-scale jets for FSRQs and LBLs (e.g., T. J. Pearson et al. 1981; S. G. Jorstad et al. 2001, 2017; M. L. Lister et al. 2009, 2013; J. J. Harwood et al. 2022; K. Yi et al. 2024; H. Shang et al. 2025; P. Thevenet et al. 2026), showing blobs emanating from a stationary core at apparent super-luminal speeds. This two-flow model has also been applied in FSRQs, the implemented work can be see in T. Vuillaume et al. (2018), the three-dimensional, relativistic, hydrodynamic simulation can be see in M.-Á. Aloy et al. (2003). However, the weak jets of HBLs assume a plume-like morphology beyond a few milliarcseconds from the core (B. G. Piner et al. 2008). The analyses above imply f_ν^{ob} of FSRQs+LBLs is more sensitive to the Doppler beaming effect than that of HBLs.

There are also the differences of spectral indices ($\alpha_r, \alpha_o, \alpha_x$) among blazars. L. Fuhrmann et al. (2016) showed that FSRQs have an average value of $\langle \alpha_r \rangle = 0.02$, BL Lacs have $\langle \alpha_r \rangle = 0.08$. R. Falomo et al. (1994) gave the $\langle \alpha_o \rangle = 0.75$, $\langle \alpha_o \rangle = 1.05$, and $\langle \alpha_o \rangle = 0.65$ for FSRQs, LBLs, and HBLs, respectively. The $\langle \alpha_x \rangle$ of FSRQs, LBLs, and HBLs are presented in Section 2.1 (i.e., 0.59, 0.92, and 1.24). But δ_i / δ_j appears to make a pivotal role in determine α_{ij} , while jet shape and α_ν play a minor role.

4.2. The Disk-Jet Connection for Blazar Sequence

The formation mechanism of relativistic jets is still controversial, but it is widely believed that the large-scale magnetic fields threading accretion disk ($\phi_B^2 \propto \dot{M}$) play a critical role in jet formation (e.g., R. D. Blandford & R. L. Znajek 1977; R. D. Blandford & D. G. Payne 1982; S. Koide et al. 2002; M. Zamaninasab et al. 2014; H. Yang et al. 2024). Therefore, it can be predicted that a more powerful accretion produces a more powerful jet, and gives rise to a larger Doppler factor. This is consistent with our observational results in Figure 3. Figure 3-(a) shows a $\log P_{\text{jet}}^{\text{in}} - \lambda_{\text{Edd}}$ correlation for our blazars, suggesting a connection between jet and disk. However, there is a break at $\lambda_{\text{Edd}} \sim -2.651$ in the $\log P_{\text{jet}}^{\text{in}} - \lambda_{\text{Edd}}$ correlation, which implies the transition of accretion mode (G. Ghisellini et al. 2011; T. Sbarrato et al. 2014). The lower accretion-rate systems (HBLs) are typically described by ADAF, which is characterized by virially hot, geometrically thick and optically thin (e.g., F. Yuan & R. Narayan 2014). The higher accretion-rate systems (FSRQs+LBLs) are usually well described by the radiatively efficient standard thin disk (N. I. Shakura & R. A. Sunyaev 1973, hereafter SSD), which consists of relatively cool, optically thick gas and is geometrically thin and optically thick. Considering the gradual evolution of accretion disk, IBLs appear to be the transitional objects between FSRQs+LBLs and HBLs. Our results are consistent with E. T. Meyer et al. (2011) and M. Keenan et al. (2021). The summary of our results can be condensed into Figure 4.

4.3. Discussion of Uncertainties in Our Analysis and Possible Alternative Interpretations

In this study, the main sources of uncertainty may arise from the representativeness of the sample, the estimation of δ_r , the empirical prescription adopted in Eq (3), the estimates of L_{disk} and M_{BH} for sources lacking direct measurements, and the extrapolation of f_x . In addition, inverse Compton cooling may also have some influence on our results. Therefore, it is necessary to briefly discuss these potential sources of uncertainty.

As shown in Section 3, the results presented in Figures 1 and 2 are consistent with those of previous studies, irrespective of the sample size adopted. This suggests that our sample could be regarded as representative of the overall blazar population. However, the number of IBLs in our sample is relatively small, and a larger IBL sample is required to further verify our results. Nevertheless, we believe that our results are reliable because previous studies based on much larger samples have shown that IBLs are distributed nearly equally between the URR and LLR (e.g., A. A. Abdo et al. 2010; J. H. Fan et al. 2016; J. H. Yang et al. 2022).

In this work, our δ_r are taken from different studies and they were estimated using different methods (see T. Hovatta et al. 2009; Z. Wu et al. 2014; I. Liodakis et al. 2017, 2018; X.-H. Ye & J.-H. Fan 2021). Additionally, δ_r from T. Hovatta et al. (2009), I. Liodakis et al. (2017) and I. Liodakis et al. (2018) are extrapolated to $\delta_{5\text{GHz}}$ using Eq (3). However, we predict this does not have a significant effect on our results, the logical explains are given as following. It is well-known that the Doppler factor of blazars is notoriously difficult to estimate due to the unobservability of β and θ . Many previous studies have proposed different methods for estimating the Doppler factor (e.g., G. Ghisellini et al. 1993; T. Hovatta et al. 2009; J.-H. Fan et al. 2014; I. Liodakis et al. 2017, 2018; X.-H. Ye & J.-H. Fan 2021; Z. Pei et al. 2022). However, the constraint of the Doppler factor of blazars is still an open question. There are some differences between the δ_r values that are estimated by different methods, but their results are strikingly consistent in showing an anti-correlation between δ_r and ν_p^{syn} (e.g., Z. Wu et al. 2007; E. Nieppola et al. 2008; W. X. Yang et al. 2022; Q.-C. Long et al. 2025), which agree with our result. For Eq (3), we consider it to be applicable for extrapolations between radio bands, as discussed in Section 2.1. Therefore, we believe that our results are robust.

The estimation of L_{disk} and M_{BH} for sources lacking direct measurements has already been discussed in Section 2.1 and Appendix A, which does not have a significant effect on our results. Most of the f_x values used in our analysis are extrapolated from other soft X-ray bands. This also does not have a significant effect on our results. Indeed, many previous studies have adopted the 1 keV X-ray band (see Section 1). However, direct measurements of the 1 keV X-ray flux are limited, and the corresponding 1 keV fluxes used in those studies were likewise extrapolated from other soft X-ray bands. Following previous work, we also extrapolated the soft X-ray fluxes to the 1 keV band. Furthermore, most of the blazars in our sample have available 0.1–2.4 keV X-ray flux measurements. We verified that using the observed 0.1–2.4 keV fluxes instead of the extrapolated 1 keV fluxes yields essentially the same results.

The blazar sequence simply refers to the decrease of ν_p^{syn} with increasing luminosity; in other words, more powerful blazars tend to have redder SED. This phenomenological trend was explained by the difference of accretion power (e.g., G. Fossati et al. 1998; M. Böttcher & C. D. Dermer 2002; J.-M. Wang et al. 2002). However, G. Ghisellini et al. (1998) suggested that blazars with lower ν_p^{syn} have a higher energy density of the external radiation field (originating from accretion disk, BLR, and dusty torus), resulting in an increasing amount of Compton cooling. As a result, the

electron energy distribution is suppressed at the high-energy end, causing the SED to shift toward lower frequencies. Compton cooling suppresses the high-energy electron population, thereby increasing the relative abundance of low-energy electrons responsible for the radio emission. This may increase $f_r^{\text{in}}/f_o^{\text{in}}$ and $f_r^{\text{in}}/f_x^{\text{in}}$, thereby providing a possible explanation for our results. However, this appears to be inconsistent with VLBA radio observations and the $\delta_r - \nu_p^{\text{syn}}$ anti-correlation. This would require that the electrons remain highly relativistic even after inverse Compton scattering. Therefore, we suggest that Compton cooling may play a role in shaping the distribution of blazars on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane, but it is unlikely to be the dominant factor.

5. SUMMARY

The bimodal distribution of blazars on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane has been found in many previous works (J. E. Ledden & S. L. O'Dell 1985; P. Padovani & P. Giommi 1995; D. Donato et al. 2001; E. Nieppola et al. 2006; A. A. Abdo et al. 2010; J. H. Fan et al. 2016; J. H. Yang et al. 2022). However, the underlying physics behind the puzzling bimodality has not yet been thoroughly investigated. In this paper, we compile a blazar sample that consists of 136 FSRQs, 64 LBLs, 39 IBLs, and 105 HBLs to explore the possible physics behind this puzzling bimodality. Our main results can be summarized as the follows:

(1) We find an anti-correlation of $\langle \delta_{5\text{GHz}} \rangle - \log \nu_p^{\text{syn}}$, which is consistent with VLBA observations. This at least implies that the jets of HBLs become sub-relativistic in the radio region, whereas the jets in strong-jet sources (FSRQs + LBLs) remain highly relativistic down to radio scales. We also find the positive correlation among $\langle \delta_{5\text{GHz}} \rangle$, $\log P_{\text{jet}}^{\text{in}}$, and λ_{Edd} for the blazar sequence (i.e., $\langle \delta_{5\text{GHz}} \rangle \propto P_{\text{jet}}^{\text{in}} \propto \lambda_{\text{Edd}}$). This suggests differences in jet strength and accretion power among blazars, where more powerful jets are preferentially produced in systems with higher accretion rates ($\phi_B^2 \propto \dot{M}$), and such stronger jets are likely to result in larger Doppler factors.

(2) The blazars show a Δ -shaped evolutionary sequence of (FSRQs+LBLs) $\rightarrow$ IBLs $\rightarrow$ HBLs on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane, with $\langle \delta_{5\text{GHz}} \rangle$, $\log P_{\text{jet}}^{\text{in}}$, and λ_{Edd} decreasing along this Δ -shaped track. FSRQs+LBLs(63/64) occupy URR of the plane with $\alpha_{\text{rx}} > 0.75$, HBLs occupy LLR of the plane with $\alpha_{\text{rx}} < 0.75$, while IBLs are almost equally distributed URR (19/39) and LLR (20/39). In addition, the distributions of α_{ox} exhibit only slight differences among blazar subclasses, whereas those of α_{ro} and α_{rx} differ significantly. Our results can be naturally explained within a framework in which the distribution of blazars on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane is regulated by the accretion rates and the Doppler beaming effect. However, the distribution of blazars appears to be more sensitive to the beaming effect than to accretion rates. Further, we find that the difference of beaming effect in different bands (δ_i/δ_j) plays a pivotal role in the distribution of blazars, the jet shape ($q = 2(3) + \alpha_\nu$) and spectral indices (α_ν) play a minor role. Our results imply the main difference of jets between FSRQs+LBLs and HBLs occur in the region from optical to radio. The jets of HBLs may begin to decelerate in the optical region and exhibit sub-relativistic speeds as they propagate to the radio region, the jets of FSRQs+LBLs are accelerated to a larger distance from the core and still maintain relativistic speeds down to the radio region. IBLs are considered transitional objects between them.

In addition, Compton cooling may play a role in shaping the distribution of blazars on the $\alpha_{\text{ro}} - \alpha_{\text{ox}}$ plane, but it is unlikely to be the dominant mechanism. Further observational and theoretical studies are needed to verify this interpretation.

(3) Our results show that, in the blazar sequence, there is a significant distinction of physical properties between FSRQs+LBLs and HBLs, IBLs appear to serve as a bridge between FSRQs+LBLs and HBLs. On average, FSRQs+LBLs are the strong-jet sources coupled with SSD, HBLs are the weak-jet sources coupled with ADAF, while IBLs are considered to be transitional objects.

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APPENDIX

A. THE ESTIMATE OF L_{DISK} AND M_{BH}

For simplicity, we present the composition of our sample in Table 3, which facilitates further analysis. There are 127/136 FSRQs, 34/64 LBLs, 8/39 IBLs, and 42/105 HBLs that have both available L_{disk} and M_{BH} . If sources without available L_{disk} or M_{BH} are eliminated, the substantial reduction of BL Lacs severely affects statistical results, especially IBLs. For example, only 8 IBLs have available L_{disk} and M_{BH} , with 6 IBLs occupying the LLR and 2 IBLs occupying the URR. However, previous works suggested that IBLs are the transitional objects between LBLs and HBLs (e.g., E. Nieppola et al. 2006; A. A. Abdo et al. 2010). Therefore, for the sources without the available L_{disk} or M_{BH} , we try to investigate a correlation between L_{disk} and other luminosities to estimate their L_{disk} , and use the fundamental plane (FP) of black hole activity to estimate their M_{BH} .

Because the broad-band emissions from AGNs are proportional to accretion rates ($f_{\nu}^{\text{in}} \propto \dot{M}^{\xi_{\nu}}$; see Section 4.1), we predict that there could be a correlation between L_{disk} and L_{ν} . We find that there is a strong correlation between L_{disk} and L_{x}^{ob} for our blazars. Similarly, we divide our sample into FSRQs+LBLs+IBL $_{>0.75}$ and HBLs+IBL $_{<0.75}$. Using the ordinary least square (OLS) bisector method (T. Isobe et al. 1990), we obtain the $L_{\text{disk}} - L_{\text{x}}^{\text{ob}}$ relation of FSRQs+LBLs+IBL $_{>0.75}$ with available L_{disk} is:

$$\log L_{\text{disk}} = (1.238 \pm 0.050) \log L_{\text{x}}^{\text{ob}} - (9.927 \pm 2.229) \quad (\text{A1})$$

with a scatter of $\sigma_L = 0.626$ dex, a Spearman correlation coefficient of $R = 0.749$ and $P = 7.639 \times 10^{-32}$. The $L_{\text{disk}} - L_{\text{x}}^{\text{ob}}$ relation of HBLs+IBL $_{<0.75}$ with available L_{disk} is:

$$\log L_{\text{disk}} = (0.901 \pm 0.098) \log L_{\text{x}}^{\text{ob}} - (3.820 \pm 4.304) \quad (\text{A2})$$

with a scatter of $\sigma_L = 0.591$ dex, a Spearman correlation coefficient of $R = 0.737$ and $P = 4.155 \times 10^{-11}$.

For the sources without available M_{BH} , we use FP to estimate their M_{BH} . The FP is a relation between radio luminosity, X-ray luminosity, and BH mass, which was firstly found by A. Merloni et al. (2003). Subsequently, FP was developed to estimate BH mass (e.g., K. Gültekin et al. 2009; L. G. C. Barriuan et al. 2022). Q.-C. Long et al. (2025) and Q.-C. Long et al. (2026) found that FSRQs+LBLs and HBLs follow different scaling relations of FP. Following K. Gültekin et al. (2009) and L. G. C. Barriuan et al. (2022), we define the FP as:

$$\log M_{\text{BH}} = \xi_{\text{mr}} \log L_{\text{r}}^{\text{ob}} + \xi_{\text{mx}} \log L_{\text{x}}^{\text{ob}} + c_0 \quad (\text{A3})$$

By using OLS bisector method, we obtain the FP of FSRQs+LBLs+IBL $_{>0.75}$ with available M_{BH} is:

$$\log M_{\text{BH}} = (0.235 \pm 0.116) \log L_{\text{r}}^{\text{ob}} + (0.334 \pm 0.140) \log L_{\text{x}}^{\text{ob}} - 16.558 \quad (\text{A4})$$

with a scatter of $\sigma_M = 0.464$ dex. The FP of HBLs+IBL $_{<0.75}$ with available M_{BH} is:

$$\log M_{\text{BH}} = (0.422 \pm 0.315) \log L_{\text{r}}^{\text{ob}} + (0.009 \pm 0.202) \log L_{\text{x}}^{\text{ob}} - 8.941 \quad (\text{A5})$$

with a scatter of $\sigma_M = 0.543$ dex.

These methods do not have a significant effect on our results. Actually, the similar methods were also used to estimate L_{disk} and M_{BH} for blazars (e.g., V. D'Elia et al. 2003; T. Sbarrato et al. 2012, 2014; Y. Sheng et al. 2024). Additionally, the $\lambda_{\text{Edd}} - \log \nu_{\text{p}}^{\text{syn}}$ relation of our sample is consistent with that of V. S. Paliya et al. (2021), which implies that our results do not have a significant bias.

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Table 3. The Composition of Our Sample

Sample	FSRQ	LBL	IBL	HBL
Tatol	136	64	39	105
L_{disk}	130	36	14	48
M_{BH}	130	48	20	83
$L_{\text{disk}} + M_{\text{BH}}$	127	34	8	42

NOTE—Total denotes the entire sample. L_{disk} denotes the subsample that consists of blazars with available L_{disk} . M_{BH} denotes the subsample that consists of blazars with available M_{BH} . $L_{\text{disk}} + M_{\text{BH}}$ denotes the subsample that consists of blazars with both available L_{disk} and M_{BH} .

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